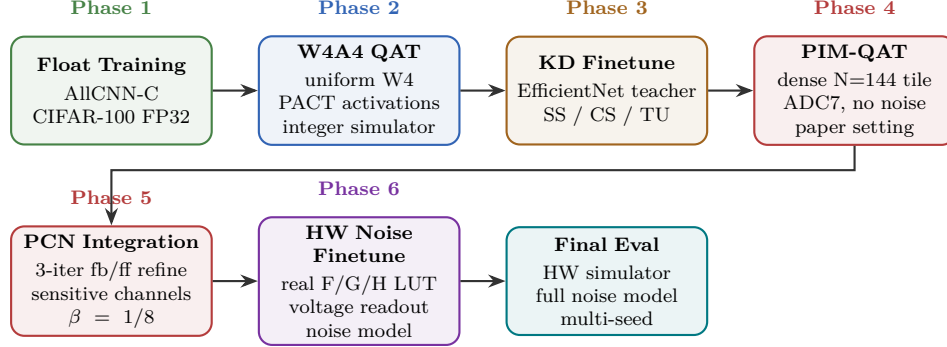


ScAN - A Path to PIM Hardware-aware Network

June 29, 2026

Phase Roadmap



Phase Results

Phase	Experiment	Precision / path	Acc. (%)	Notes
1	RGB AllCNN-C + BN	FP32 float	72.99	Standard CIFAR-100 RGB, $3 \times 32 \times 32$, crop/flip augmentation.
1	RAW/RGGB AllCNN-C	FP32 float	61.31	Current 4-channel RGGB BN baseline, raw_noise=True .
2	RAW/RGGB W4A4 SS	uniform W4A4 INT sim	56.61	Main noisy self-studying baseline; use uniform path going forward.
2	RAW/RGGB W4A4 SS + PACT	uniform I16/W4/A4 INT sim	60.01	Uniform path with LUT-Q disabled; gain comes from PACT/I16 and simulator alignment.
3	QKD CS + PACT	I16/W4/A4 LUT-Q $m = 15$	61.39	Non-PIM reference; integer LUT weights are not the uniform $[-7, 7]$ PIM target.
3	QKD TU + PACT	I16/W4/A4 LUT-Q $m = 15$	61.82	Non-PIM reference; best dense int4 simulator result, not used as the PIM baseline.
4	PIM-QAT dense N144	paper W4/A4/ADC7 simulator	56.75	Self-study from FP32, uniform W4 + PACT, 7-bit ADC, N=144, no noise; GPU forward best was 56.88.
4	PIM-QAT dense N144, Phase 2 init	paper W4/A4/ADC7 simulator	58.80	Active uniform PIM baseline; initialized from Phase 2 uniform I16/W4/A4 + PACT checkpoint.
4	PIM-QKD CS controlled	paper W4/A4/ADC7 simulator	58.28	Started from the active PIM baseline; best CS full-simulator result did not improve over Phase 2 init.
4	PIM-QKD TU controlled	paper W4/A4/ADC7 simulator	58.13	Started from controlled PIM-CS; GPU forward reached 58.75, but full simulator remained below baseline.

Model / stage	Acc. (%)	Notes
EfficientNetV2-L teacher, clean	77.30	Verified on current H5 test split.
EfficientNetV2-L teacher, noisy	77.33	Default noisy evaluation.
Co-studied teacher, noisy	78.70	Earlier CS run improved teacher.
Uniform SS + PACT rerun	60.01	I16/W4/A4, <code>qkd_use_lutq=False</code> , fixed integer simulator.
QKD CS rerun	61.39	I16/W4/A4 LUT-Q $m = 15$ + PACT; non-PIM reference.
QKD TU rerun	61.82	Best non-PIM dense int4 simulator reference.
PIM-QKD CS, controlled	58.28	Full PIM simulator; uniform $[-7, 7]$, A4, I16 input, ADC7/N144.
PIM-QKD TU, controlled	58.13	Full PIM simulator; started from controlled PIM-CS.

Item	Current setting
Model	ALL-CNN-C, 4-channel RGGB input, BatchNorm; GBN and PCN disabled for the minimal pass.
Weights	Uniform 4-bit with deployed integer code range $[-7, \dots, 7]$ and precomputed per-output-channel scales.
Activations	First layer input is unsigned I16; hidden activations are unsigned A4 codes in $[0, 15]$ and values in $[0, 1]$.
Tile	Hidden PIM layers use dense N=144 tiles. Stage1.0 and logits stay digital for this pass.
ADC	7-bit paper PIM quantizer for bit-serial partial sums, forward scale 1.03, no read noise.
Simulator	Full test uses the PIM simulator with integer weights, integer activations, precomputed scales, and no runtime <code>tanh</code> .

Teacher / KD

Current PIM Setting

Next

1. Use the 58.80% dense N144 PIM-QAT checkpoint as the current minimal uniform PIM baseline.
2. Keep QKD results separated by constraint: LUT-Q $m = 15$ is a non-PIM reference; uniform $[-7, 7]$ PIM-QKD did not improve this run.
3. Add the `pim_info/hardware_model.tex` voltage-domain readout once F/G/H and noise parameters are fixed.
4. Add PCN only after the minimal BN PIM path is stable.